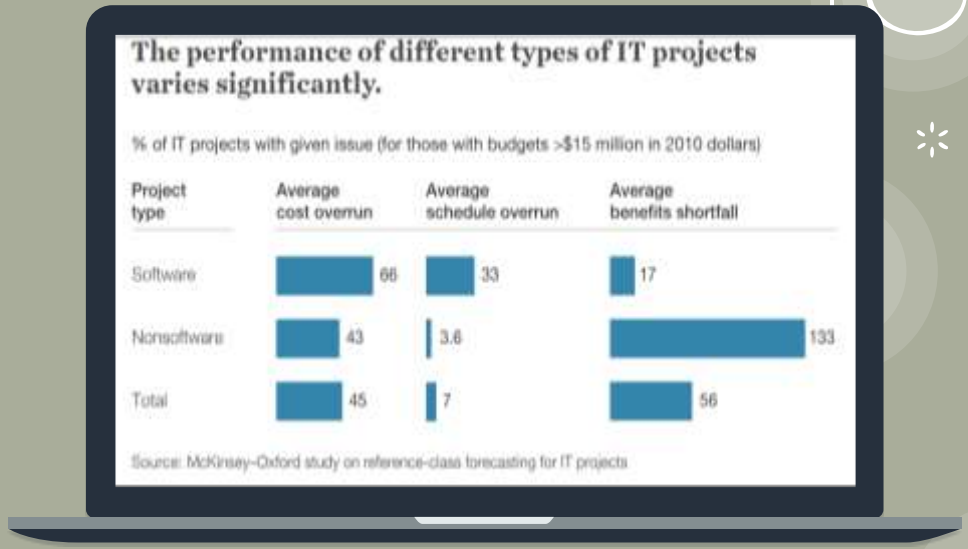
Risk Management

Anything that can go wrong will go wrong

Murphy's law

IT Software Project

No project is risk-proof, and IT projects are no exception. There are various risks in IT projects from clients frequently changing their requirements to overshooting budgets and deadlines



Risk management

1. Identification

Identify risks that may interfere with project objectives.

2. Analysis

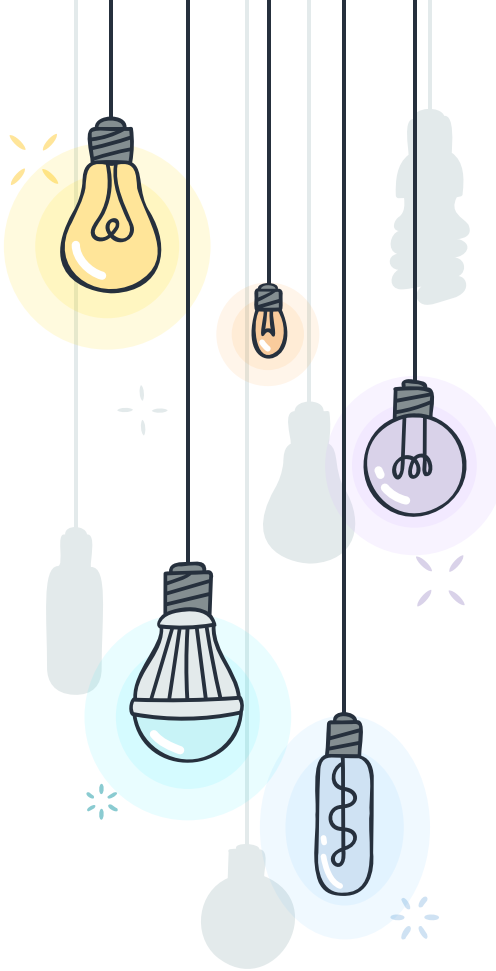
Determine which of the identified risks are the most dangerous. For each risk, we will evaluate its Probability and Impact on some scale proposed by experts.

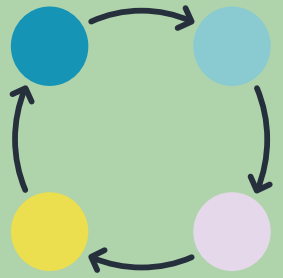
3. Planning

Project management step. For each risk, from the list of critical ones, it is necessary to come up with a strategy that will protect our project from it.

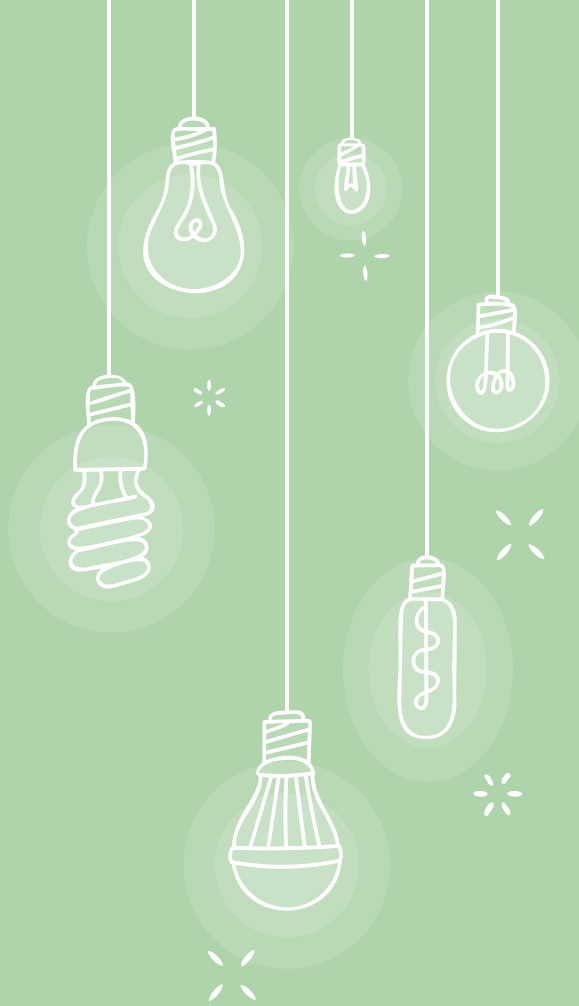
4. Monitoring and control

This is more of a process than a step. Its goal is to keep the risk list and project plan up to date.





Risk management is an ongoing process. Throughout the life of a project a risk management team are always identifying new risks and reassessing existing ones



All of the software, tools and management elixirs in the world will not save project management. Only people can save project management.

Risk Managemet Team

The Risk Manager

The central figure for the project's risk management activities. Provides an overview of the project's risks as well as their mitigation plan. Defining project costs by assessing risks, opportunities, and actions. Supporting risk owners in characterizing risks and their associated cost. Reporting: creating indicators, updating and communicating to project stakeholders

The Risk Owner

The risk owner is responsible for each risk. They work alongside the risk manager as they hold all the technical information on risks. They can be at the origin of its identification and therefore characterize it by defining the cause or causes of the risk and its consequences. Then, they make sure to have the updated information on their risks.

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The Project Manager

The project manager works in collaboration with the risk manager. They set expectations and validate the proposals made by the risk manager on the process and status for their project. They must have a global vision of the risks. Their role is also to position themselves on the critical risk response strategies proposed by risk owners, for example to accept a plan to reduce probability or impact.

The Project Team

The risk manager will meet with the project team to discuss the possible impacts on the project's schedule and budget. Project team can identify risks

The Sponsor

The sponsor can define a global budget for risks thanks to the first information provided by the Risk Manager. During the life of the project, they will be required to validate this budget (cost of risks and associated actions) or to release the funds for confirmed risks.

Delphi method

A good way to hear all the team members. Not only the loud voices

Delphi method

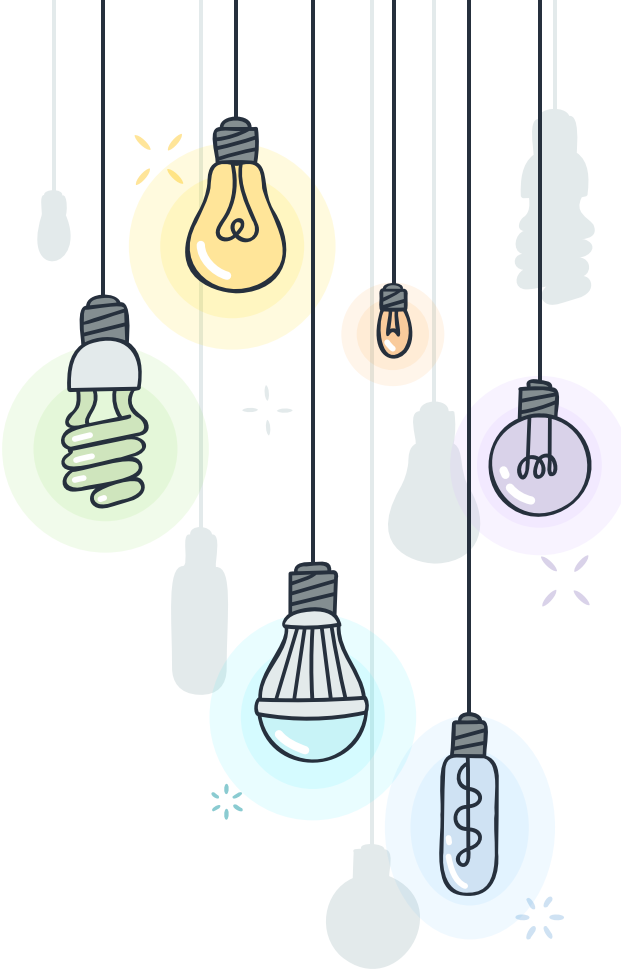
Coordinator asks for input all team specialists, while keeping their identities anonymous.

Then these opinions are send to the members for review.

Thus the opinions of all the participants carry the same weight

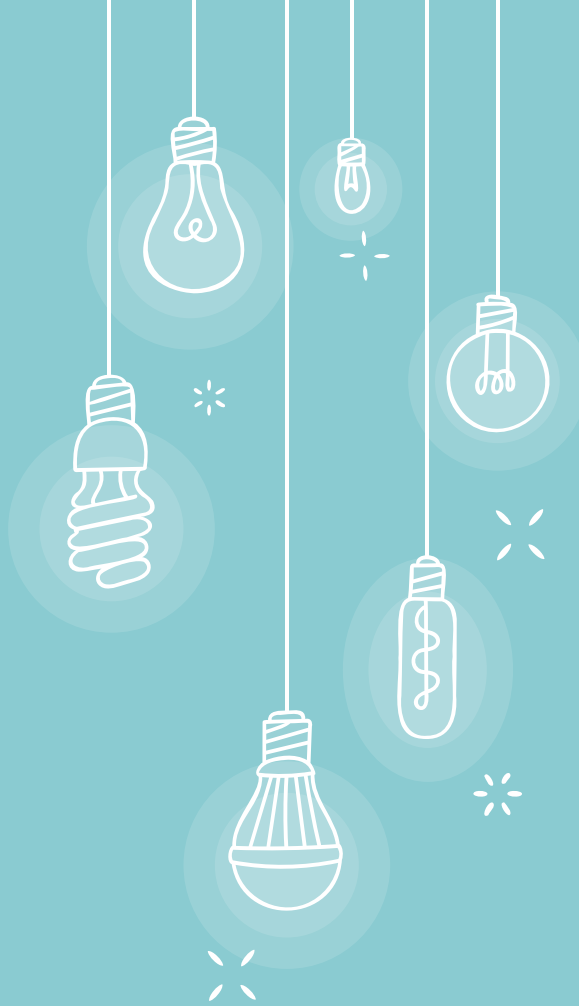
Risk Breakdown Structure

- IT Project
  - Business
    - Competitors
    - Suppliers
    - Cash Flow
  - Technical
    - Hardware
    - Software
    - Network
  - Organizational
    - Executive Support
    - User Support
    - Team Support
  - Project Management
    - Estimates
    - Communication
    - Resources



Risk statement :

“If event “X” occurs, then project objective “Y” will be affected”



Risk Register of an IT Software project example

#	Risk Statement	Probability (1-10)	Imapct (1-10)	Prob*Imp (1-100)
1	The customer often does not respond for several days. We may not have time to approve the specifications for the next phase of the project in time. Start of development will be delayed by a few days (behind schedule)	8	7	56
2	There are a lot of tickets in the bug tracking tool with outdated status. We can deal with non-critical bugs that are marked as critical, and vice versa, miss important ones. There will be loss of time and quality of the product	9	7	64
3	Developers from the customer's side also participate in the work on the project. They can make significant changes to the code and architecture of the project without warning us. Time will be spent discussing potential bugs and possibly rollbacks to previous versions of the project.	8	8	64
4	Manual testing of the project takes more and more time with each passed phase. Over time, we may end up spending weeks of manual labor. We will receive delays in the delivery of the project and quality problems.	8	9	72
5	When the current version of the product is already in production. Urgent changes coming from end users may affect our current plans. We will be late with the development of a new phase of the project	9	8	72
6	The earthquake will destroy the office	1	10	10

Numerical Risk Matrix

Risk Impact	Risk Probability 1	2	3	4	5	6	7	8	9	10
10	Risk 6 10									
9								Risk 4 72		
8								Risk 3 64	Risk 5 72	
7								Risk 1 56	Risk 2 64	
6										
5										
4										
3										
2										
1										

Strategies

1. **Accept the Risk**
   Understanding that the risk may occur. Having a plan for what to do if it does occur, but do nit take any actions to prevent the risk

2. **Avoid the Risk**
   Changing our plans so that we eliminate the risk (try to prevent from occuring, trying to make its probability as low as possiple)

3. **Transfer the risk**
   Making another party responsible. Transfer the risk to a party who should be more qualified to deal with it

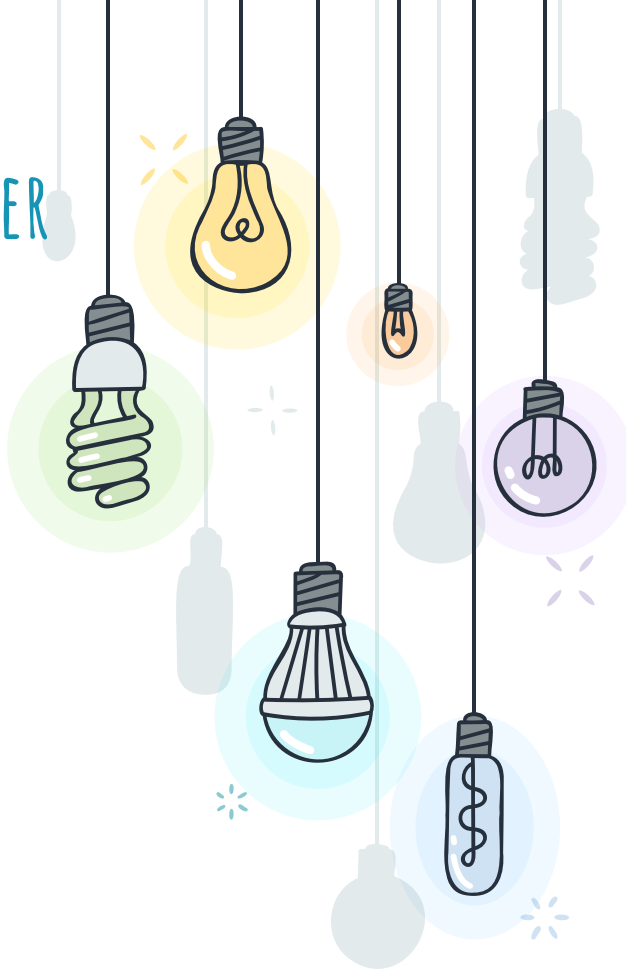
4. **Mitigate the Risk**
   Taking action to make the risk less impactful

Risk Register

#	Risk Statement	Strategy	Owner	Detailed Plan
1	The customer often does not respond for several days. We may not have time to approve the specifications for the next phase of the project in time. Start of development will be delayed by a few days (behind schedule)	Transfer	employee name	Notify the customer that he is responsible for any delays caused by his absence
2	There are a lot of tickets in the bug tracking tool with outdated status. We can deal with non-critical bugs that are marked as critical, and vice versa, miss important ones. There will be loss of time and quality of the product	Avoid	employee name	Assign a responsible person for updating and checking statuses
3	Developers from the customer's side also participate in the work on the project. They can make significant changes to the code and architecture of the project without warning us. Time will be spent discussing potential bugs and possibly rollbacks to previous versions of the project.	Transfer	employee name	Notify the customer that he is responsible for any delays caused by his developers
4	Manual testing of the project takes more and more time with each passed phase. Over time, we may end up spending weeks of manual labor. We will receive delays in the delivery of the project and quality problems.	Avoid	employee name	Add a task to the project plan to study the auto-testing tool and implement it in our processes
5	When the current version of the product is already in production. Urgent changes coming from end users may affect our current plans. We will be late with the development of a new phase of the project	Accept	employee name	Add five days of reserve to the project plan
6	The earthquake will destroy the office	Accept		Our prayers

Some other things that can be in Risk Register

- Symptoms and warning signs (“triggers”) of risks’ actual imminent occurrence
- Budget and schedule activities required to implement the chosen responses
- Contingency reserves of time and/or money
- Risk owner assigned responsibilities



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Thanks!

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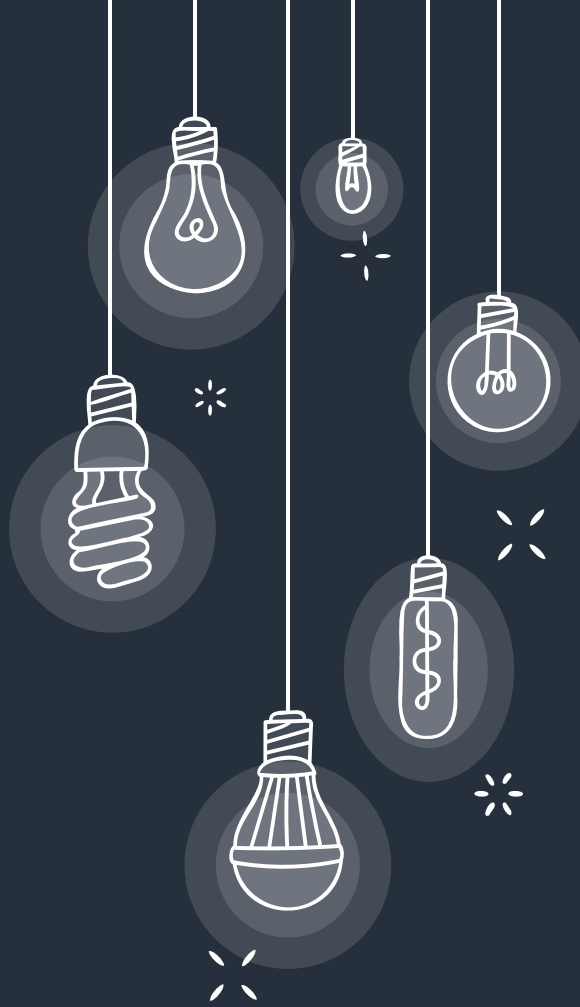

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